



Agenda

- ① Why do we need distributed system
- ② Basics of Distributed System
- ③ challenges of Distributed system
- ④ Architecture of Hadoop Framework

Why do we need distributed system

1994 Amazon / Relentless

① sells Books

② Only sold in USA

Product	Customer Details	Customer Purchase
① ID	① Customer ID	① Order ID
② Thumbnail *	② Customer Address	② Product ID
③ Title	③ Payment Details *	③ Customer ID
④ Description	④ phone	④ Order Date
⑤ Price	⑤ Email	⑤ Status
⑥ Category	⑥ profile photo	
50.1 KB	200 Bytes	100 Bytes

Basic calculation

Images ≈ 50 KB

Other details ≈ 100 Bytes

10^3 Product + 100 customer + 10^4 customer purchase $\approx$

$$50 \times 10^3 \times 10^2 + 100 \times 200 + 10^4 \times 100 = 50 \text{ MB}$$

Imagining 50 GB space $\rightarrow$ 9000 + days

2025 - Amazon

Product

Extensive metadata
High Resolution Image
Product Videos

Customer Details

Session Storage

Customer Purchase

GPS Data for
tracking



Approximate Data Footprint for 2025

Video $\approx 50\text{MB}$

High Quality Image $\approx 500\text{KB}$

Product +

10,000

customer +

1000

customer purchase $\approx$

2.87 million

With increase in complexity and increase in volume and velocity storing 1TB per day per region is impossible in single machines

Dilemma: Vertical vs Horizontal Scaling

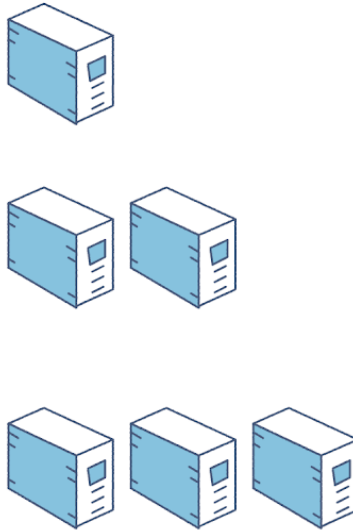
Vertical Scaling

add more resources to the same server



Horizontal Scaling

add more servers to scale



① Scalability

VS: will not work

HS: Forced for HS

② Fault tolerance

VS: SPOF

HS: reducing risk of SPOF

③ Performance:

VS: with ever increasing volume, VS gives up after a point

HS: can be tweaked to be performant.

④ Cost Effectiveness:

VS: 1 Bq large machine

HS: lot of smaller machines

↳ you can scale up & scale down according to traffic

Challenges of Horizontal Scaling

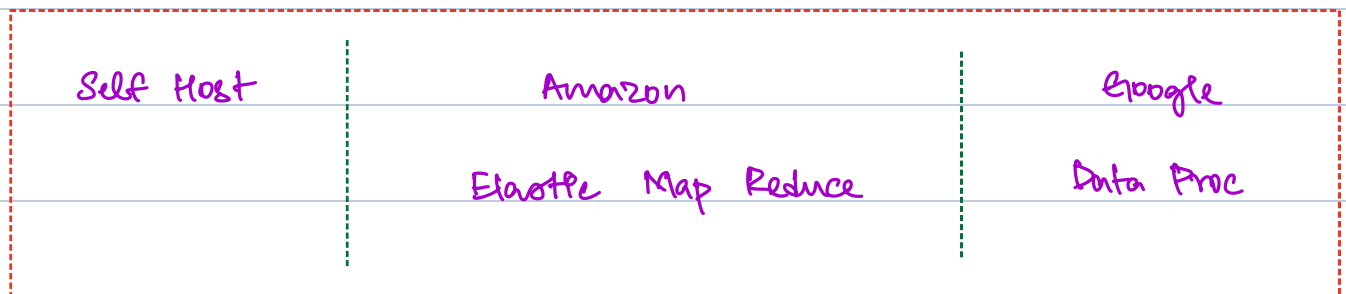
- ① Data Consistency
- ② Configuration and management
- ③ System Failure
- ④ Limited Bandwidth

Hadoop Framework — MEAN (MongoDB, Express, Angular, Node)

Hadoop is a framework that allows distributed processing of large dataset across clusters of commodity computers using simple programming models

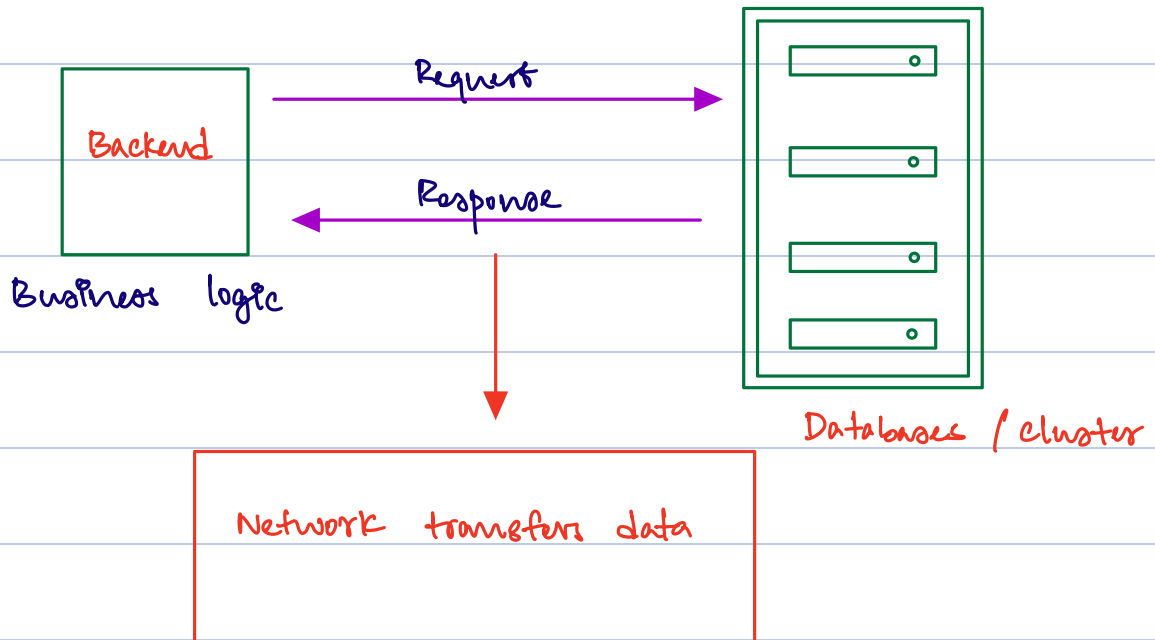
Characteristics of Hadoop Framework

- ① Scalable
- ② Reliable
- ③ Economical
- ④ Maintainable



Hadoop Framework

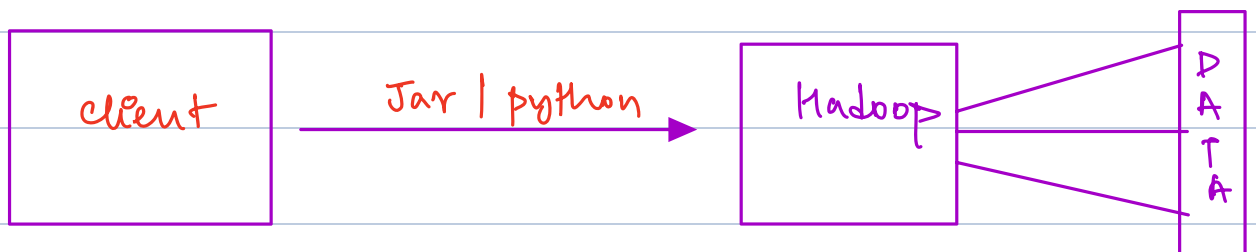
Traditional Systems



Pizza - Data

Hands - Network

Mouth - Business processing



Hadoop System



Food - Data

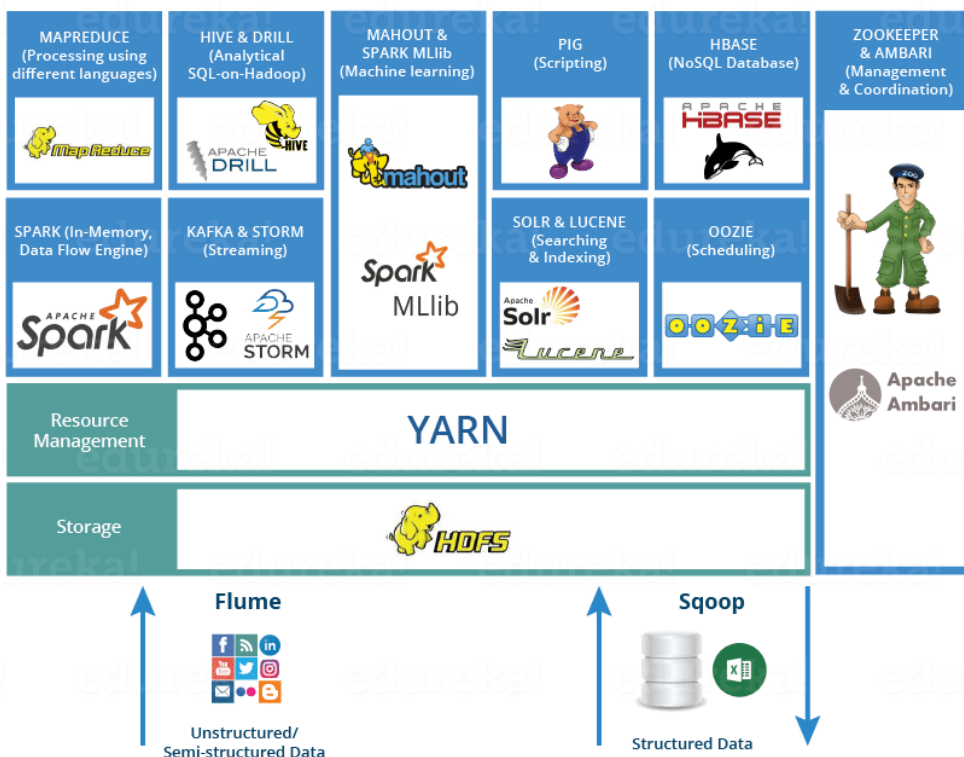
Month - Business logic

very minimal use of
Hands

Use Case

Understand the log data available across multiple servers
to get the hour with the least load to apply a patch

Hadoop framework components



This Week

- ① HDFS
- ② YARN
- ③ Map
- Reduce

10:01 → 10:10pm

HDFS - Hadoop Distributed File System

Distribute File system that provides access to data across
hadoop cluster

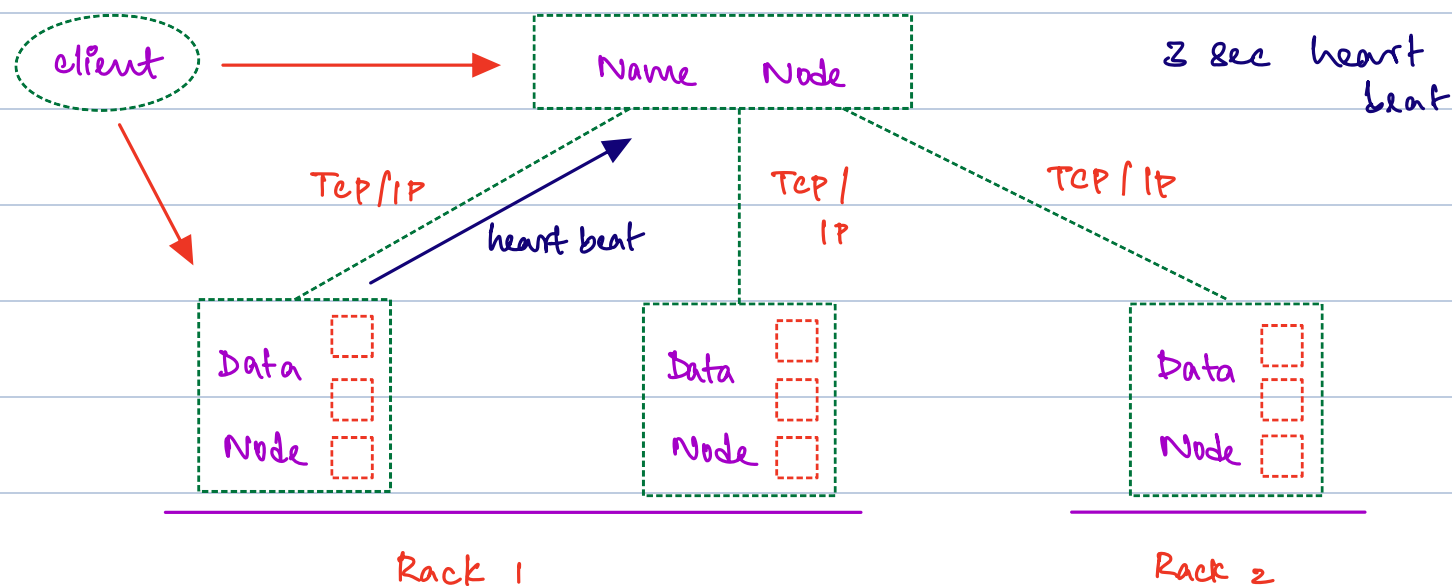
Windows

C:/ D:/

Linux

/ /user /opt
/lib

HDFS Architecture



Name Node

① Service which runs inside a Master Node

② Stores Metadata

① Directory structure

② File permission, ownership

③ File Block location, file name

③ Keeps track of Data Nodes Health.

④ Maintains Rack Awareness

① File creation, deletion, rename

② Directory operation

③ permission changes

event log

FS

Image

Data Node

- ① It stores data.
- ② Handles all requests on files such as reading/writing data into HDFS

File Blocks — will be stored in your data node

HDFS automatically splits the files into file blocks of 128 MB each.

- ① we store and replicate these blocks

128	128	128	116
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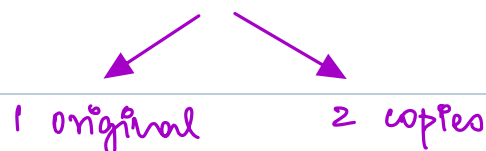
500 MB

- ② Split Data → helps distributed computing
- ③ Storage Efficiency

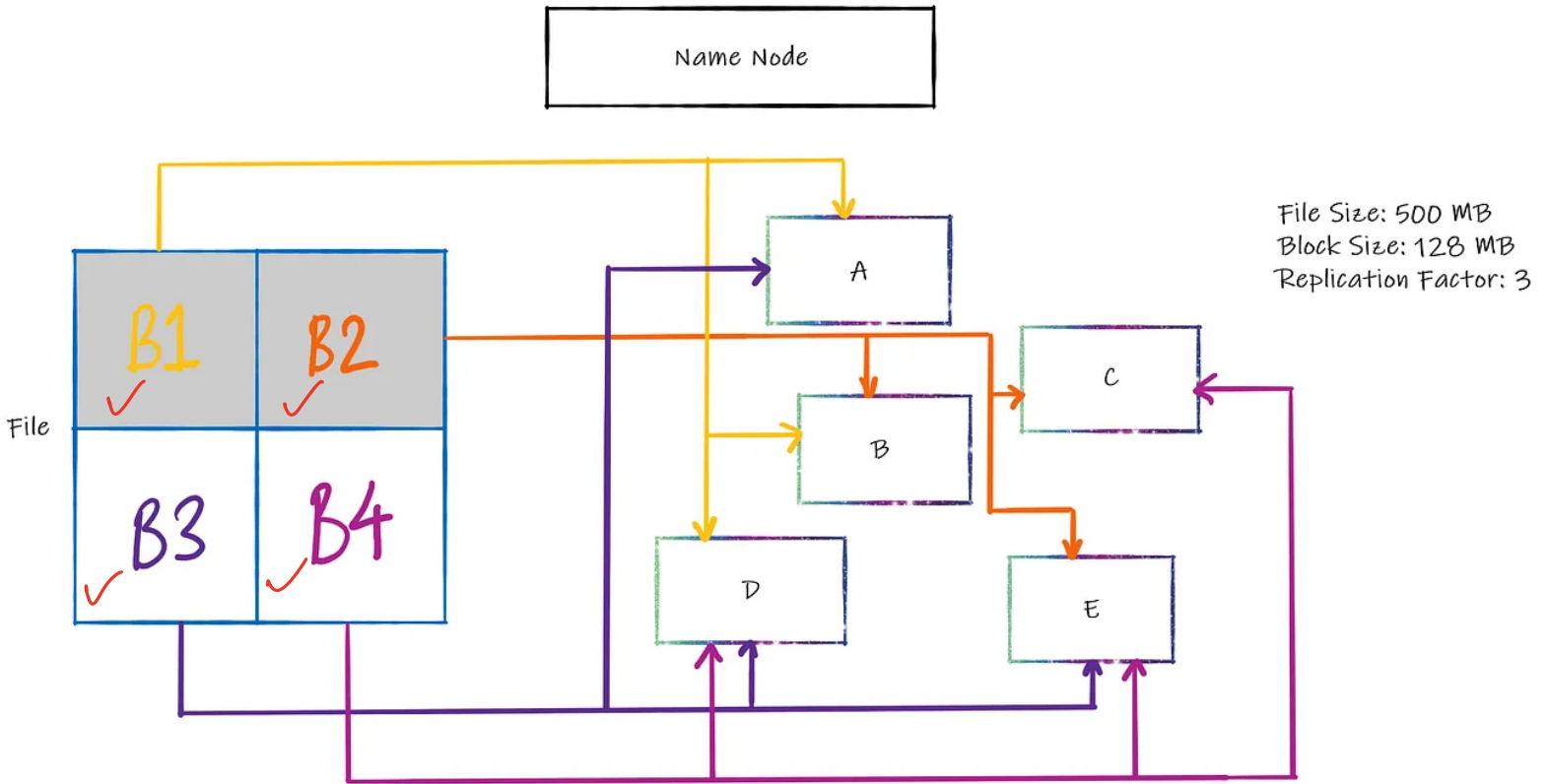
Replication Factor — 3 Data File x 3 time

For a resilient system, we have a replication mechanism in HDFS which based on its configuration will make copies of file blocks and store them in data-nodes

Default Replication : 3

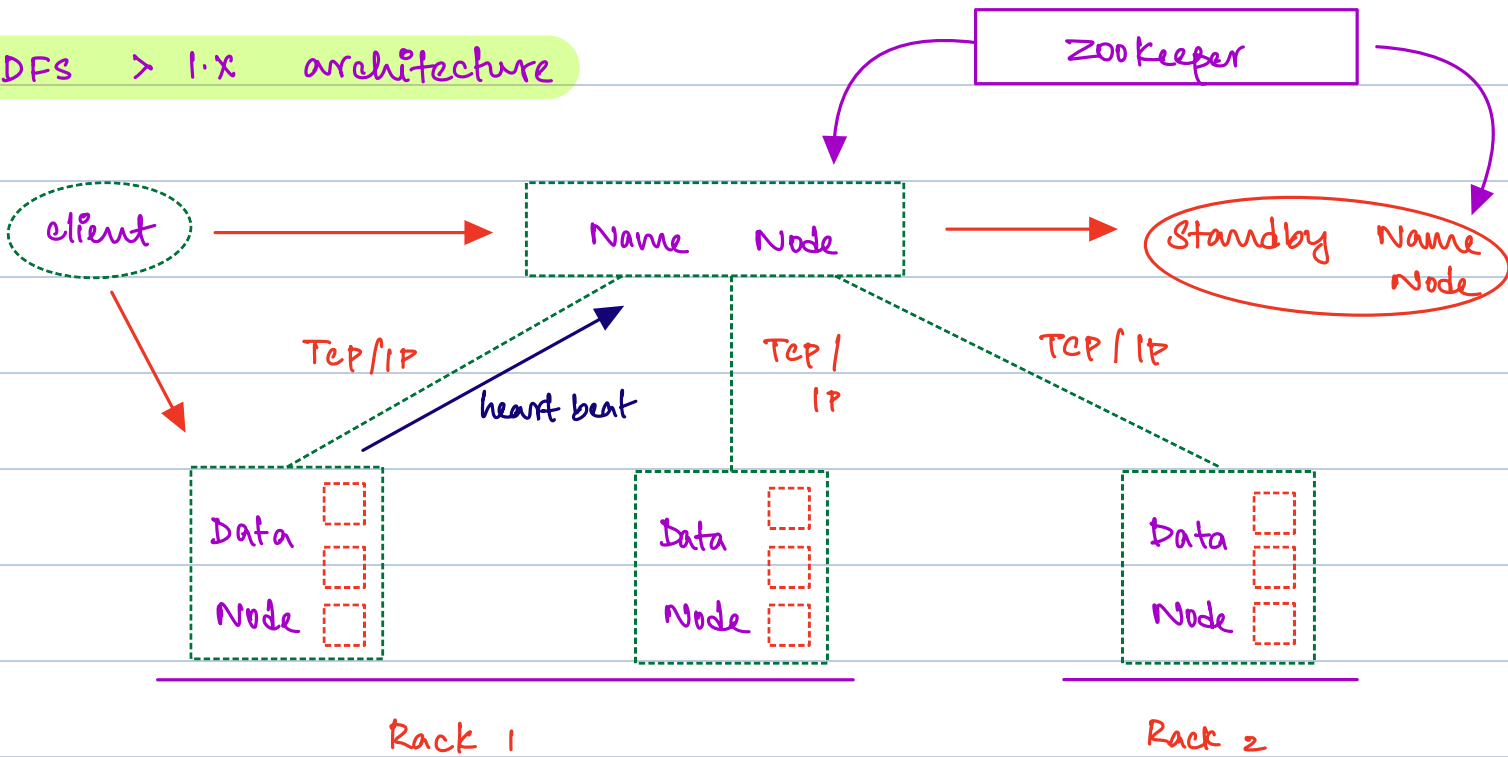


HDFS Storage and Replication



A, B, C, D, E are data nodes

HDFS > 1.x architecture



Rack awareness

topology → node that are part of one rack

Next class

- ① Flow of Reads and writes
- ② HDFS commands and YARN
- ③ YARN